Customers Like You

“Customers who bought this also bought…” and customer groups

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Data Engineer: from 1.07M lines to a clean matrix

1,067,371

raw order lines, Dec 2009 to Dec 2011

684,000

clean lines used for training (to 8 Oct 2011)

5,471 x 3,686

customers x products, 98% empty

- Removed: lines with no customer ID (about 23%), cancellations, zero or negative quantity or price, non-product codes, two fully reversed orders. Every rule is logged.
- Purchase value = log(1 + quantity), so a few huge wholesale lines cannot dominate. Products bought by fewer than 10 customers are dropped.
- Time-based split: train, then 2 months validation (tuning), then the last 2 months as test. Held-out = products a customer never bought before. Leak checks are asserted in code.
- One command, python build_matrix.py, rebuilds everything from the raw file.

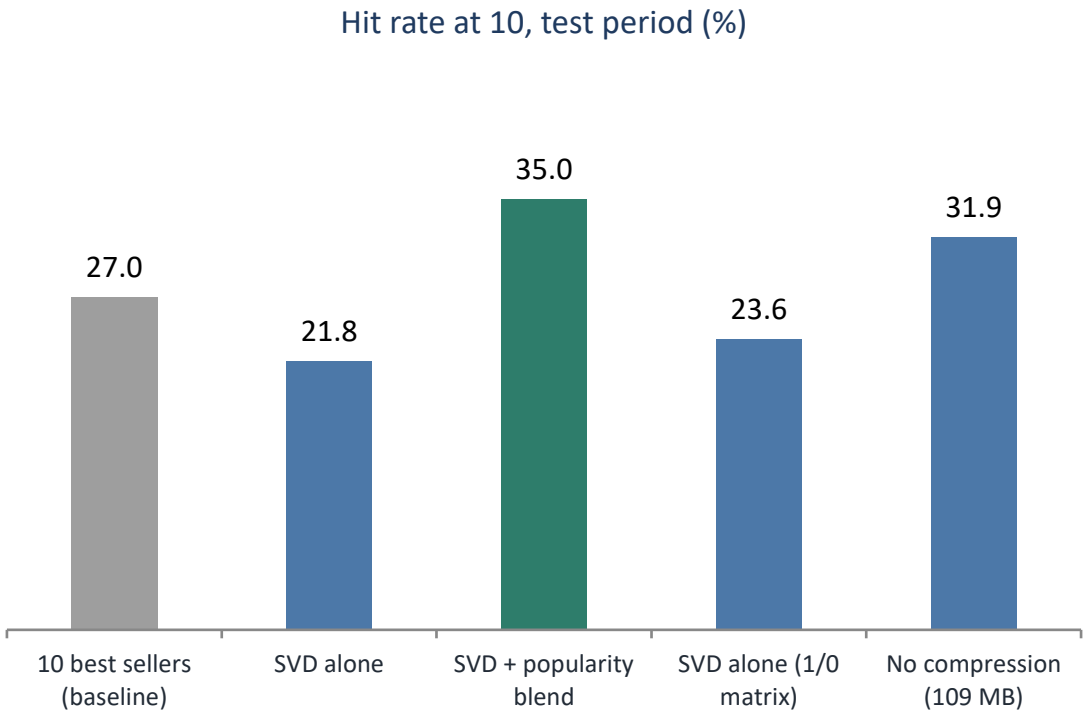
Method: Truncated SVD, chosen before fitting

- Why SVD: the table is 98% empty and SVD works on it directly. Customer and product factors come from one fit, which is what a recommender needs.
- PCA centres the data, so it describes deviation from the average customer. I predicted this would change what component 1 means, and wrote it down before running anything.
- Settings: 50 shopping patterns (k = 50) and a popularity blend (weight 2), both chosen on the validation period only. The test period was scored once.
- Result of the PCA vs SVD check: same error, similar fit time, and component 1 is overall volume and popularity in both. My prediction was wrong, and the log says so.

PCA vs SVD at k = 50

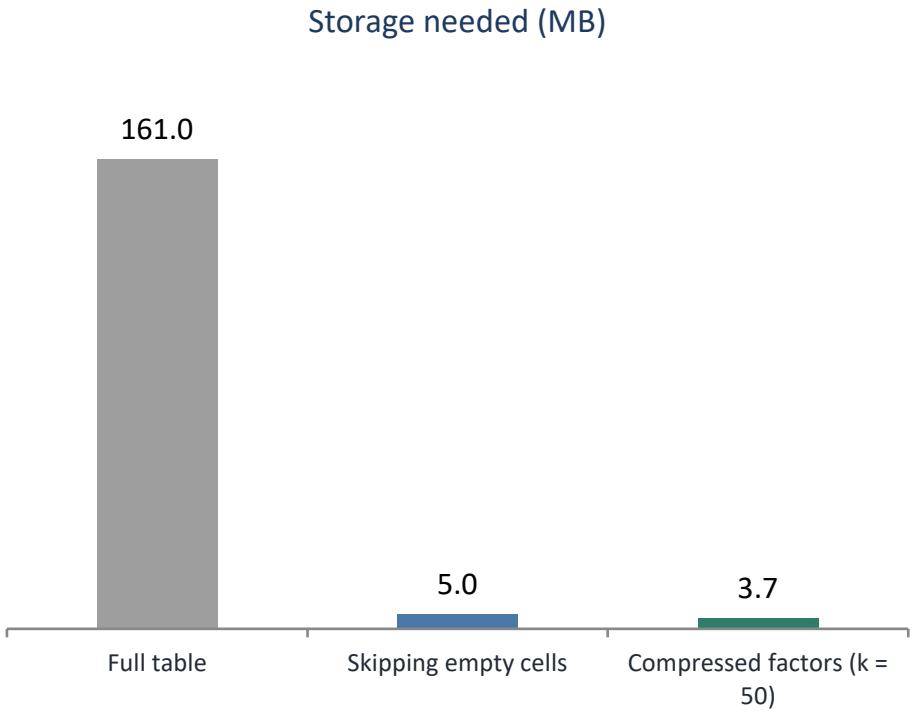
- Reconstruction error: a tie
- Fit time: about a second each
- Hit rate: SVD 21.8%, PCA 24.6% with a centred profile (extra check run after seeing test scores)
- Both below the 27.0% baseline

Does it work? Only the blend beats best sellers



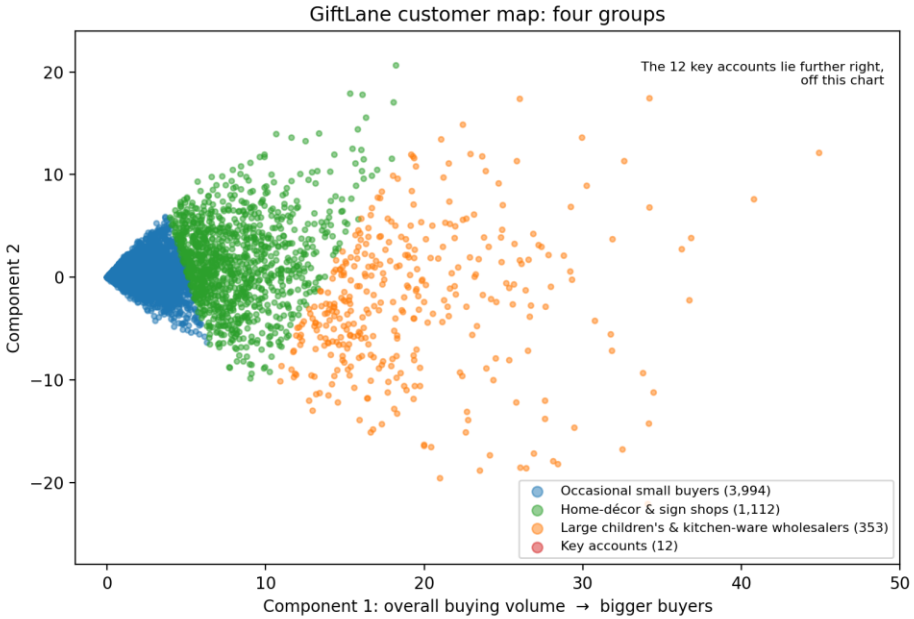
- Hit rate = share of customers with at least one of 10 suggestions that they then bought for the first time. Margin about ±1 point.
- Plain SVD is below best sellers (21.8% vs 27.0%). The blend adds +8.0 points (±0.9).
- Validation said 34.5%, test says 35.0%, so the tuning did not overfit.
- The 109 MB uncompressed table is not a fair comparison with the blend: it has no popularity boost.

Size and speed: the nightly job takes about 2 seconds



- Nightly job: fit, suggestions for every customer and an “also bought” list for every product, about 2 seconds (1.3 to 2.3 across runs). The target was under 60.
- Factors are 3.7 MB against 161 MB dense, about 98% smaller.
- Most of the saving comes from skipping empty cells (161 to 5 MB), not from compression itself.
- What is lost: 50 patterns keep about 38% of the detail, so broad habits, not one-off choices.

Customer map: four groups marketing can name



Group	Customers	Revenue
Occasional small buyers	73%	23%
Home-décor and sign shops	20%	34%
Children's and kitchen-ware wholesalers	6.5%	31%
Key accounts (12 customers)	0.2%	12.6%

Groups differ mostly in how much they buy, less in what they like. Revenue is to 8 Oct 2011, before returns, rounded. Only group names appear: no customer IDs.

What could go wrong, and what next

Risks

- New customers get no personal suggestions (380 of 2,422 in the test period).
- Products with under 10 buyers are left out; the 23% of lines with no customer ID were unusable.
- Test window was the Christmas run-up; other months may differ.
- The blend leans to best sellers; variety was not measured.
- 12 customers bring 12.6% of revenue.

Next steps

- A/B test the blend on a few product pages.
- Give the uncompressed table the same blend, to measure the true cost of compressing.
- Score the product-page lists, which were only checked by eye.
- Refresh nightly; re-check across seasons.

Recommendation: pilot the blend on a few live product pages.